STAT 2020 Discussion
Week 6

Shuo Li

The University of Iowa
Department of Statistics and Actuarial Science

February 23, 2026

Index (Outline)

1. Admin & Reminders
2. Key Distributions
3. Practice (Identification + Range)
4. Homework Note
5. Wrap-Up
6. Attendance Check

Key Skill for Quizzes/Exams

- On future exams/quizzes, you may not be told which distribution to use.
- You should be able to:
  - identify the correct distribution,
  - state the **range** of X,
  - compute PMF, mean, and variance.

How to Identify the Distribution (Quick Flow)

- **Fixed number of trials n?**
  - Yes $\Rightarrow$ Binomial (with replacement / independent) or Hypergeometric (without replacement)
  - No $\Rightarrow$ Geometric / Negative Binomial (waiting time for successes)
- **Without replacement?** $\Rightarrow$ Hypergeometric
- **Stop when you reach r successes?** $\Rightarrow$ Negative Binomial
- **Stop when you reach 1 success?** $\Rightarrow$ Geometric

Geometric Distribution

Sequence of independent Bernoulli trials with success probability p.
Let X = trial on which we get the **first** success.

$$X \sim \mathrm{Geom}(p).$$

Geometric Distribution

Sequence of independent Bernoulli trials with success probability p.
Let X = trial on which we get the **first** success.

$$X \sim \text{Geom}(p).$$

$$P(X = x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots$$

Geometric Distribution

Sequence of independent Bernoulli trials with success probability p.
Let X = trial on which we get the **first** success.

$$X \sim \text{Geom}(p).$$

$$P(X = x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots$$

$$E(X) = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2}.$$

From Geometric to Negative Binomial

Geometric = waiting time for **1** success.
Negative Binomial = waiting time for r **successes**.

From Geometric to Negative Binomial

Geometric = waiting time for **1** success.
Negative Binomial = waiting time for r **successes**.
Let

$$X_1, X_2, \ldots, X_r \sim \mathrm{Geom}(p)$$

be independent waiting times between successive successes.

From Geometric to Negative Binomial

Geometric = waiting time for **1** success.
Negative Binomial = waiting time for r **successes**.
Let

$$X_1, X_2, \ldots, X_r \sim \mathrm{Geom}(p)$$

be independent waiting times between successive successes.
Then the total trial number of the r-th success is

$$X = X_1 + X_2 + \cdots + X_r.$$

From Geometric to Negative Binomial

Geometric = waiting time for **1** success.
Negative Binomial = waiting time for r **successes**.
Let

$$X_1, X_2, \ldots, X_r \sim \mathrm{Geom}(p)$$

be independent waiting times between successive successes.
Then the total trial number of the r-th success is

$$X = X_1 + X_2 + \cdots + X_r.$$

$$\boxed{\text{Negative Binomial} = \text{sum of } r \text{ Geometric variables}}$$

Negative Binomial Distribution

Let X = trial on which we get the r-th success.

$$X \sim \mathrm{NB}(r, p).$$

Negative Binomial Distribution

Let X = trial on which we get the r-th success.

$$X \sim \mathrm{NB}(r, p).$$

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \ldots$$

Negative Binomial Distribution

Let X = trial on which we get the r-th success.

$$X \sim \mathrm{NB}(r, p).$$

$$P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \ldots$$

$$E(X) = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

Analogy: Bernoulli vs Binomial

Fixed number of trials n:

- Bernoulli = outcome of **one** trial
- Binomial = total successes in n **trials**

Analogy: Bernoulli vs Binomial

Fixed number of trials n:

- Bernoulli = outcome of **one** trial
- Binomial = total successes in n **trials**

Waiting for successes:

- Geometric = waiting time for **1** success
- Negative Binomial = waiting time for r **successes**

Analogy: Bernoulli vs Binomial

Fixed number of trials n:

- Bernoulli = outcome of **one** trial
- Binomial = total successes in n **trials**

Waiting for successes:

- Geometric = waiting time for **1** success
- Negative Binomial = waiting time for r **successes**

$$\text{Binomial} = \text{sum of Bernoulli}$$

$$\text{Negative Binomial} = \text{sum of Geometric}$$

Hypergeometric

Population size N, with M successes and $N - M$ failures. Randomly select n objects **without replacement**. Let $X =$ number of successes selected.

$$X \sim \mathrm{HG}(n, N, M).$$

Hypergeometric

Population size N, with M successes and $N - M$ failures. Randomly select n objects **without replacement**. Let $X =$ number of successes selected.

$$X \sim \mathrm{HG}(n, N, M).$$

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}.$$

Hypergeometric

Population size N, with M successes and $N - M$ failures. Randomly select n objects **without replacement**. Let $X =$ number of successes selected.

$$X \sim \mathrm{HG}(n, N, M).$$

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}.$$

Range:

$$\max\{0,\ n - (N - M)\} \leq x \leq \min\{n,\ M\}.$$

Hypergeometric: Mean and Variance

$$E(X) = n\frac{M}{N}.$$

$$\mathrm{Var}(X) = n\frac{M}{N}\left(1 - \frac{M}{N}\right)\frac{N-n}{N-1}.$$

Hypergeometric: Mean and Variance

$$E(X) = n\frac{M}{N}.$$

$$\mathrm{Var}(X) = n\frac{M}{N}\left(1 - \frac{M}{N}\right)\frac{N-n}{N-1}.$$

- Note: if N is large and n is small, then HG can be approximated by Binomial with $p = M/N$.

Practice 1 (NB): Identify + Range

A player makes independent shots. Each shot is a hit with probability $p = 0.4$.
Let X = shot number when the player gets the **3rd** hit.

- Identify the distribution of X
- What is the range of X?

Practice 1 (NB): Identify + Range

A player makes independent shots. Each shot is a hit with probability $p = 0.4$.
Let X = shot number when the player gets the **3rd** hit.

- Identify the distribution of X
- What is the range of X?

Answer:

$$X \sim \mathrm{NB}(r = 3, p = 0.4), \quad x = 3, 4, 5, \ldots$$

Practice 2 (Geom): Simple PMF

Each attempt succeeds with probability $p = 0.25$. Attempts are independent.
Let X = attempt number of the **first** success.

$$X \sim \mathrm{Geom}(0.25).$$

Practice 2 (Geom): Simple PMF

Each attempt succeeds with probability $p = 0.25$. Attempts are independent.
Let X = attempt number of the **first** success.

$$X \sim \mathrm{Geom}(0.25).$$

Compute $P(X = 1)$ and $P(X = 2)$.

Practice 2 (Geom): Simple PMF

Each attempt succeeds with probability $p = 0.25$. Attempts are independent.
Let X = attempt number of the **first** success.

$$X \sim \mathrm{Geom}(0.25).$$

Compute $P(X = 1)$ and $P(X = 2)$.

$$P(X = 1) = 0.25, \qquad P(X = 2) = 0.25(0.75).$$

Practice 3 (HG): Expectation + Range

A class has 20 students, including 5 statistics majors.
Select 4 students **without replacement**.
Let X = number of statistics majors selected.

- Identify the distribution of X
- Find the possible values of X
- Compute $E(X)$

Practice 3 (HG): Expectation + Range

A class has 20 students, including 5 statistics majors.
Select 4 students **without replacement**.
Let X = number of statistics majors selected.

- Identify the distribution of X
- Find the possible values of X
- Compute $E(X)$

Answer:

$$X \sim \mathrm{HG}(n = 4, N = 20, M = 5).$$

Range:

$$0 \leq x \leq \min\{4, 5\} = 4.$$

$$E(X) = 4 \cdot \frac{5}{20} = 1.$$

HW 4(d): Full Setup

Given:

- Each throw hits bullseye with probability $p = 0.3$
- Tyler throws until he gets 2 bullseyes
- Let $X =$ number of throws needed

HW 4(d): Full Setup

Given:

- Each throw hits bullseye with probability $p = 0.3$
- Tyler throws until he gets 2 bullseyes
- Let $X =$ number of throws needed

Part (d):

- Suppose he is stopped after 6 throws (even if he has fewer than 2 bullseyes)
- Let $Y = \min(X, 6)$
- Compare $E(Y)$ with 4

HW 4(d): Strategy

Recall:

$$X \sim \mathrm{NB}(2, 0.3)$$

He is stopped after 6 throws.

$$Y = \min(X, 6).$$

HW 4(d): Strategy

Recall:

$$X \sim \mathrm{NB}(2, 0.3)$$

He is stopped after 6 throws.

$$Y = \min(X, 6).$$

Compute expectation directly:

$$E(Y) = \sum_{x=2}^{5} xP(X = x) + 6P(X \geq 6).$$

HW 4(d): Strategy

Recall:

$$X \sim \mathrm{NB}(2, 0.3)$$

He is stopped after 6 throws.

$$Y = \min(X, 6).$$

Compute expectation directly:

$$E(Y) = \sum_{x=2}^{5} xP(X = x) + 6P(X \geq 6).$$

Idea:

- For $X = 2, 3, 4, 5$, $Y = X$
- For $X \geq 6$, $Y = 6$

Wrap-Up: Key Takeaways

- **NB / Geom**: independent trials, count **trial number** when success count reaches r
  - Geom $=$ NB with $r = 1$
  - Range: Geom starts at 1; NB starts at r
- **HG**: sample **without replacement**
  - PMF uses combinations
  - Range: $\max\{0, n - (N - M)\} \leq x \leq \min\{n, M\}$

In-Class Attendance Check (Quick)

Suppose $X \sim \mathrm{Geom}(p)$ using the definition:

$$P(X = x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots$$

If $p = 0.30$, compute $P(X = 1)$.

Please submit: Full Name; Section Number; Your Answer

In-Class Attendance Check (Quick)

Suppose $X \sim \mathrm{Geom}(p)$ using the definition:

$$P(X = x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots$$

If $p = 0.30$, compute $P(X = 1)$.

Please submit: Full Name; Section Number; Your Answer

Answer:

$$P(X = 1) = 0.30.$$